

Baseline Model Comparison Report

This report summarizes the performance of 9 baseline models evaluated on the Autism Facial Expression Recognition dataset. The objective of this phase was to determine how well standard convolutional and transformer-based architectures can classify 6 distinct emotions (anger, fear, joy, natural, sadness, surprise) directly from raw images, establishing a benchmark for future architectural improvements.

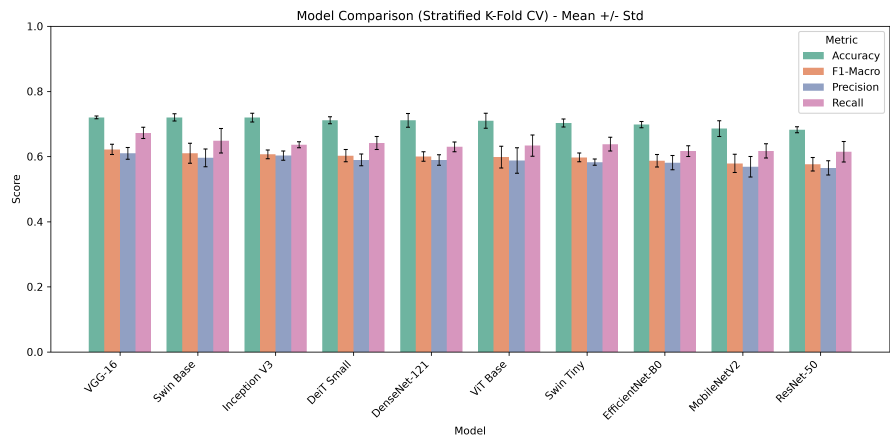
Summary Table

Model	Accuracy	F1-Macro	Precision	Recall
VGG-16	0.7201	0.6217	0.6098	0.6725
Swin Base	0.7201	0.6100	0.5961	0.6485
Inception V3	0.7196	0.6067	0.6030	0.6363
DeiT Small	0.7113	0.6026	0.5895	0.6417
DenseNet-121	0.7113	0.6001	0.5894	0.6298
ViT Base	0.7102	0.5984	0.5880	0.6339
Swin Tiny	0.7030	0.5972	0.5827	0.6381
EfficientNet-B0	0.6980	0.5871	0.5813	0.6167
MobileNetV2	0.6858	0.5790	0.5687	0.6174
ResNet-50	0.6825	0.5762	0.5652	0.6150

Note: All models were evaluated using Stratified 5-Fold Cross-Validation. The metrics presented represent the average performance across all five folds, ensuring a robust and statistically sound evaluation.

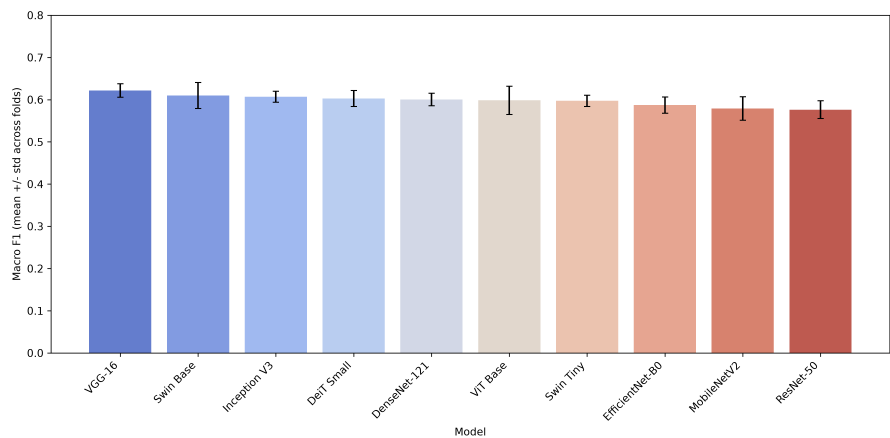
Key Charts and Takeaways

1. Overall Performance



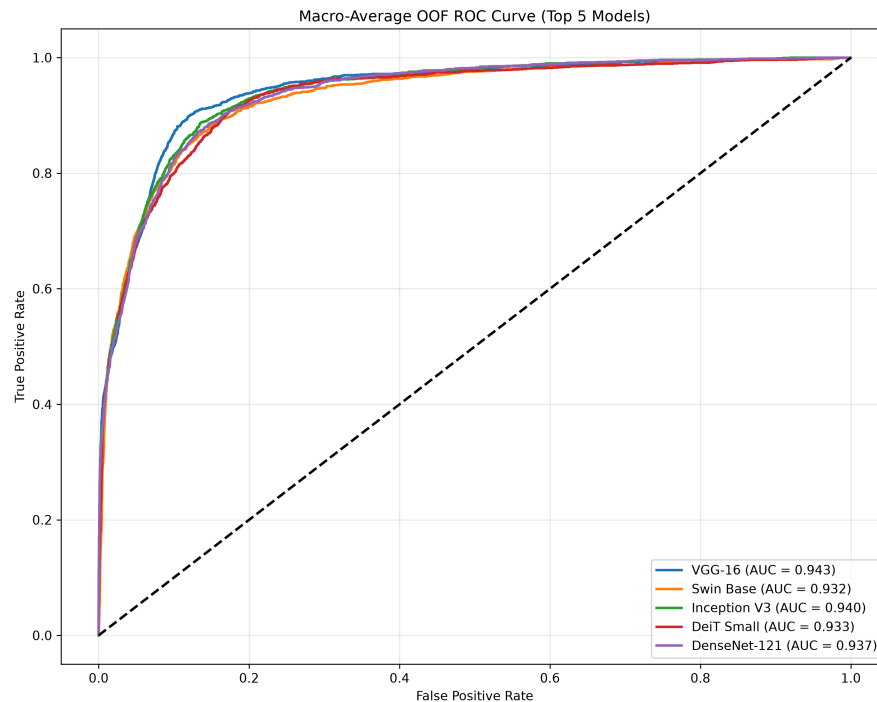
Takeaway: VGG-16 and Swin Base achieved the highest overall performance across the evaluated metrics, making them the strongest baseline models.

2. Balancing Rare Emotions



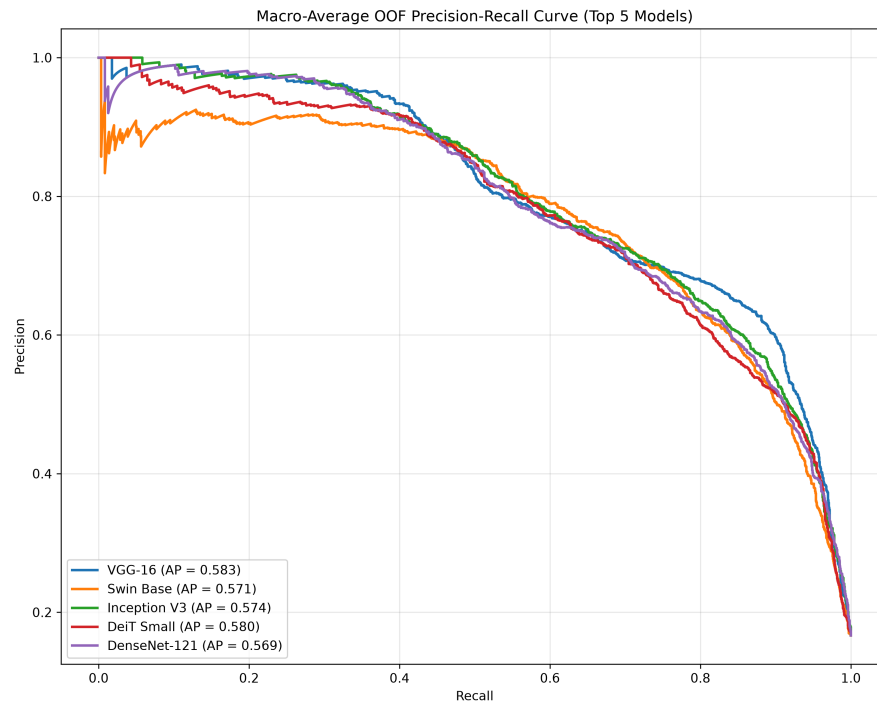
Takeaway: The Macro-F1 score indicates how effectively a model manages class imbalance rather than defaulting to majority classes. VGG-16 demonstrates the most consistent ability to balance performance across all emotion categories.

3. ROC Curve (Top 5 Models)



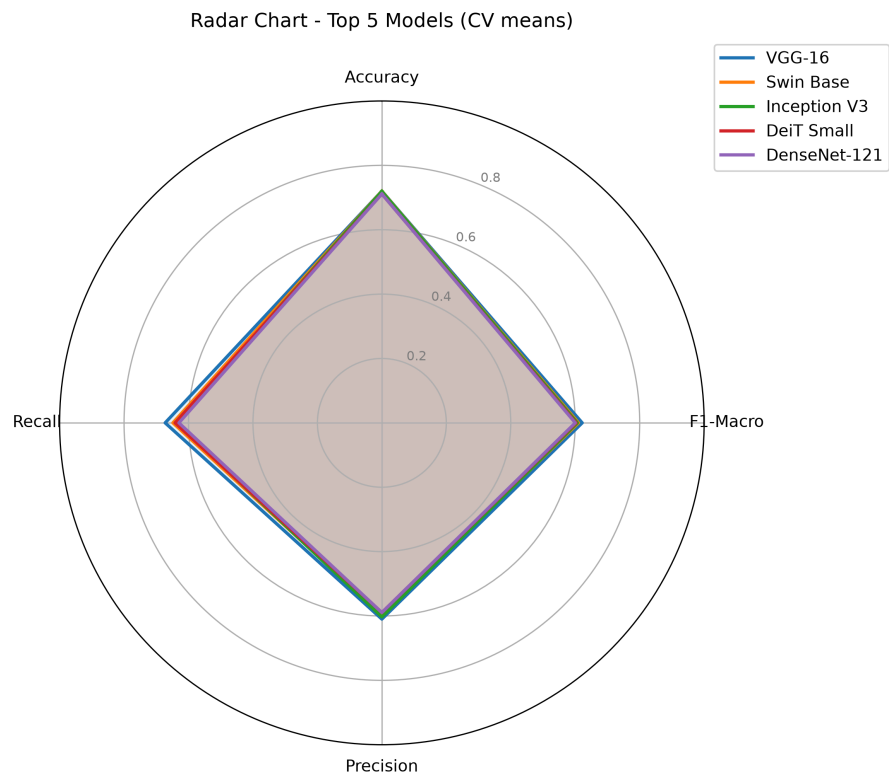
Takeaway: The top 5 models achieve high Area Under the Curve (AUC) scores, demonstrating that the raw, unaugmented image data contains sufficient discriminatory features for accurate classification.

4. Precision vs Recall



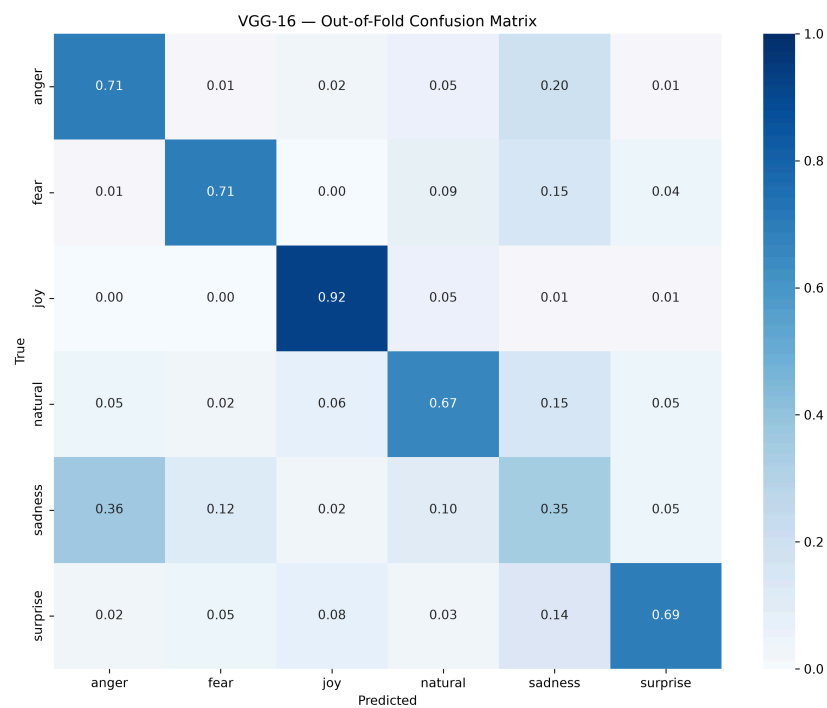
Takeaway: The tight clustering among the leading models indicates a strong capacity to maintain high precision without sacrificing recall, which is essential for handling minority classes.

5. Radar Chart



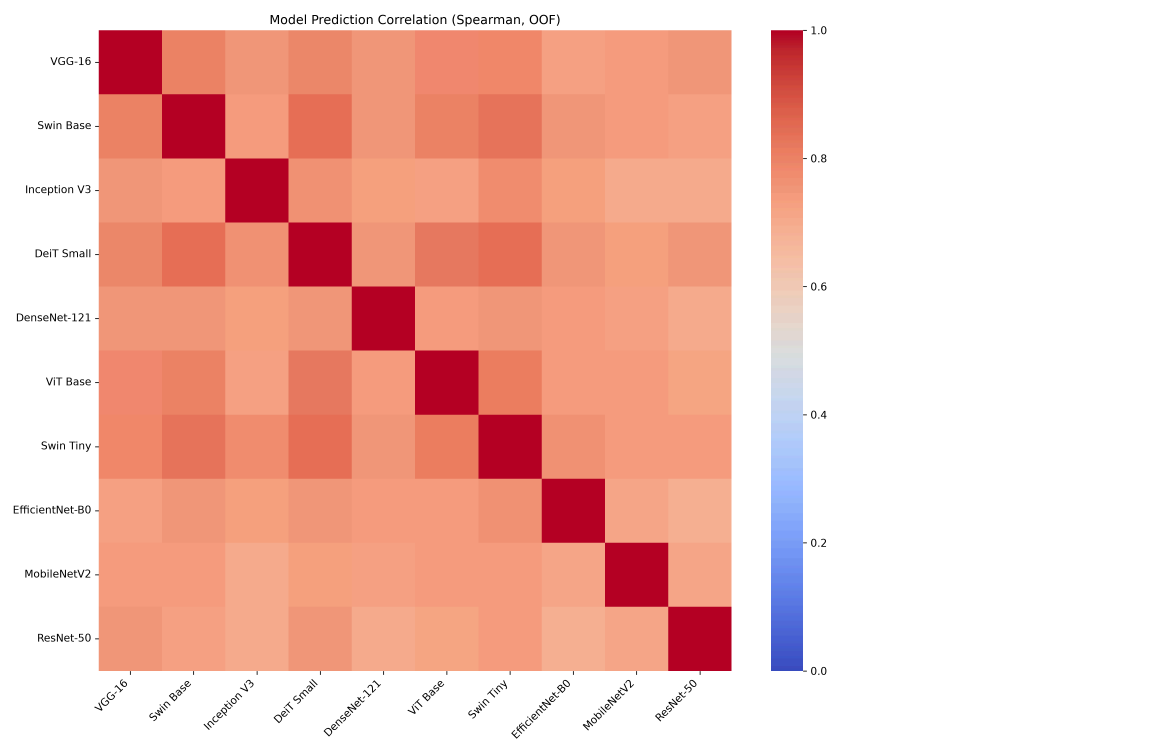
Takeaway: The multi-metric footprint visually confirms that VGG-16 exhibits well-rounded performance, without isolated failures in any single evaluation metric.

6. Where the Best Model Struggles



Takeaway: The confusion matrix for VGG-16 reveals that the model still occasionally misclassifies “Fear” and “Surprise.” This is an expected limitation due to the relative scarcity of training samples for these specific expressions.

7. Feature Correlation Analysis



Takeaway: The correlation heatmap indicates that CNN- and Transformer-based models capture different aspects of facial features. This observation may help guide the design of an improved architecture in the next phase of the research.

Methodology Overview

To ensure the evaluation was fair and representative of real-world generalization: 1. **Stratified 5-Fold Cross-Validation:** Instead of using a fixed test split, the 1,808 images were merged and divided into 5 folds. The models trained on 4 parts and tested on the remaining 1 part, repeating 5 times. This ensures every image is tested exactly once, producing highly reliable metrics. 2. **Handling Class Imbalance:** During training, minority classes were sampled more frequently so that the model learned all emotion classes more effectively.

Conclusion

- **VGG-16** achieved the best overall performance among all baseline models.
- **Swin Base** produced comparable accuracy, demonstrating the effectiveness of Transformer-based models.

- All baseline models achieved approximately 68-72% accuracy, indicating that further improvements require architectural innovation rather than simply changing the backbone.
- Based on these findings, the next stage of this research is to develop a proposed dual-stream architecture by combining CNN and Transformer features.